

By Theorem 2, since there are no rows of the matrix in the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$, where $b \neq 0$, the system is consistent, i.e. there exists a solution set.

$\therefore b$ is a linear combination of a_1, a_2 and a_3 . $\square$

$$12. a_1 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}, a_2 = \begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix}, a_3 = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}, b = \begin{bmatrix} -5 \\ 11 \\ -7 \end{bmatrix}$$

Solution: We need to determine whether there exists a solution to the equation $x_1 a_1 + x_2 a_2 + x_3 a_3 = b$, for some $x_1, x_2, x_3 \in \mathbb{R}$.

This is equivalent to finding whether the augmented matrix $[a_1 \ a_2 \ a_3 \ b]$ has a solution set.

$$\begin{bmatrix} 1 & 0 & 2 & -5 \\ -2 & 5 & 0 & 11 \\ 2 & 5 & 8 & -7 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 2 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3} - 2 \times \text{Row 1}} \begin{bmatrix} 1 & 0 & 2 & -5 \\ 0 & 5 & 4 & 1 \\ 0 & 5 & 4 & -7 \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ - \text{Row 2}}]{\text{Row 3} = \text{Row 3} - \text{Row 2}} \begin{bmatrix} 1 & 0 & 2 & -5 \\ 0 & 5 & 4 & 1 \\ 0 & 0 & 0 & -8 \end{bmatrix}$$

Verification of whether final matrix is in echelon form

1. There are no zero rows in the matrix. $\therefore$ The condition that all the non-zero rows of the matrix must be above the zero rows of the matrix is vacuously True.

2. Row 1 Leading Entry = 1 in Column 1
 Row 2 Leading Entry = 5 in Column 2
 Row 3 Leading Entry = 2 in Column 4

$\therefore$ The condition that the Leading Entry of a row must be in a column to the right of the columns of the Leading Entries of all such rows above it is True.